

February Crossword Solutions

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1 1	2 8
3 3	7

ONE ACROSS

There are $(2n)!$ total permutations of the $2n$ distinct people. For each couple $i \in \{1, 2, \dots, n\}$, let E_i be the event that the two members of couple i stand next to each other. We seek

$$a_n = (2n)! - \left| \bigcup_{i=1}^n E_i \right|.$$

Fix a subset $S \subseteq \{1, 2, \dots, n\}$ with $|S| = k$. Count the permutations in which every couple in S is adjacent. For each $i \in S$, glue the two people of couple i into a single block. Then the lineup consists of $(2n - k)$ objects: the k glued blocks together with the remaining $2n - 2k$ individuals. These $(2n - k)$ objects can be arranged in $(2n - k)!$ ways, and each glued block has 2 internal orders, giving an additional factor of 2^k . Hence

$$\left| \bigcap_{i \in S} E_i \right| = 2^k (2n - k)!.$$

There are $\binom{n}{k}$ choices for S , so by inclusion–exclusion,

$$a_n = \sum_{k=0}^n (-1)^k \binom{n}{k} 2^k (2n - k)!.$$

For asymptotics, divide by $(2n)!$:

$$\frac{a_n}{(2n)!} = \sum_{k=0}^n (-1)^k \binom{n}{k} \frac{2^k}{(2n)_k}, \quad (2n)_k = (2n)(2n-1) \cdots (2n-k+1).$$

For each fixed k as $n \rightarrow \infty$,

$$\binom{n}{k} \sim \frac{n^k}{k!} \quad \text{and} \quad (2n)_k \sim (2n)^k,$$

so

$$\binom{n}{k} \frac{2^k}{(2n)_k} \sim \frac{n^k}{k!} \cdot \frac{2^k}{(2n)^k} = \frac{1}{k!}.$$

Therefore the sum converges to

$$\lim_{n \rightarrow \infty} \frac{a_n}{(2n)!} = \sum_{k=0}^{\infty} (-1)^k \frac{1}{k!} = e^{-1}.$$

Equivalently,

$$a_n \sim \frac{(2n)!}{e} \quad (n \rightarrow \infty).$$

Thus, $c = e$, and since $e = 2.718\dots$, the sum of the first four digits is then $2 + 7 + 1 + 8 = \boxed{18}$.

ONE DOWN

The given system resembles the real and imaginary parts of the expansion of $(x + iy)^3$. Recall the binomial expansion:

$$(x + iy)^3 = x^3 + 3x^2(iy) + 3x(iy)^2 + (iy)^3 = (x^3 - 3xy^2) + i(3x^2y - y^3).$$

We are given $x^3 - 3xy^2 = 46$ and $y^3 - 3x^2y = 9$, which implies $3x^2y - y^3 = -9$. Let $z = x + iy$. Substituting the given values into the expansion, we have:

$$z^3 = 46 - 9i.$$

We seek $S = x^2 + y^2$, which is exactly the squared magnitude of the complex number z , denoted as $|z|^2$. First, we compute the squared magnitude of z^3 :

$$|z^3|^2 = 46^2 + (-9)^2 = 2116 + 81 = 2197.$$

We observe that $2197 = 13^3$. Using the property $|z^3| = |z|^3$, we have:

$$(|z|^3)^2 = 13^3 \implies |z|^6 = 13^3.$$

Taking the cube root of both sides gives $|z|^2 = 13$. Thus, $S = x^2 + y^2 = \boxed{13}$.

TWO DOWN

Let O be the center of the circle, and let P be any point inside the circle with $OP = r > 0$. Take any diameter; it lies on some line ℓ passing through O . The perpendicular distance from P to the line ℓ is

$$d(P, \ell) = r \sin \theta$$

for some angle θ , so in particular $0 \leq d(P, \ell) \leq r$ (since $|\sin \theta| \leq 1$).

But the problem's condition requires that $OP = r$ be less than the perpendicular distance from P to *every* diameter line, i.e.

$$r < d(P, \ell)$$

for each diameter line ℓ . This is impossible because always $d(P, \ell) \leq r$.

Therefore, no point P in the disk satisfies the condition, so the described region is empty. Hence its area is 0, making $A + 87 = \boxed{87}$.

THREE ACROSS

We wish to compute $S \pmod{2026}$. First, we determine the prime factorization of the modulus: $2026 = 2 \times 1013$. We can verify that 1013 is prime (testing primes up to $\sqrt{1013} \approx 31.8$ shows no factors). By the Chinese Remainder Theorem, it suffices to find $S \pmod{2}$ and $S \pmod{1013}$.

$$S = \sum_{k=1}^{2025} k^{2025} \equiv \sum_{k=1}^{2025} k \pmod{2}.$$

In the range 1 to 2025, there are 1013 odd numbers and 1012 even numbers. The sum of 1012 even numbers is 0 (mod 2), and the sum of 1013 odd numbers is 1 (mod 2). Thus $S \equiv 1 \pmod{2}$. By Fermat's Little Theorem, $k^{1012} \equiv 1 \pmod{1013}$ for $k \not\equiv 0$. The exponent is $2025 = 2 \cdot 1012 + 1$. Therefore:

$$k^{2025} = (k^{1012})^2 \cdot k^1 \equiv 1^2 \cdot k \equiv k \pmod{1013}.$$

This holds for all k . The sum simplifies to:

$$S \equiv \sum_{k=1}^{2025} k \pmod{1013}.$$

The terms 1 to 1013 sum to a multiple of 1013. The remaining terms 1014 to 2025 correspond to residues 1 through 1012. The sum of 1 to 1012 is $\frac{1012 \cdot 1013}{2} = 506 \cdot 1013 \equiv 0 \pmod{1013}$. Thus $S \equiv 0 \pmod{1013}$. We seek $R \in \{0, \dots, 2025\}$ such that $R \equiv 1 \pmod{2}$ and $R \equiv 0 \pmod{1013}$. The multiples of 1013 are 0, 1013, 2026. 0 is even; 1013 is odd. Thus $R = 1013$. Then, $R^1 \equiv 13^1 \equiv (13^3)^3 \times 13^2 \equiv (-3)^3 \cdot 69 \equiv -1863 \equiv \boxed{37} \pmod{100}$.